AMMUNITION AFTER 1900

Following the development of the unitary brass cartridge, which combined all three essential elements (primer, propellant, and projectile) in one package, it only remained for the nature of those elements to be improved. Primers became more effective and bullets became more aerodynamic and capable of accuracy at long ranges. However, the most important developments were in propellant. In the final decade of the 19th century, propellants evolved, with the advent of smokeless powder and later of a nitroglycerine-based mixture generally known as cordite. This replaced gunpowder entirely.

Rifle cartridges

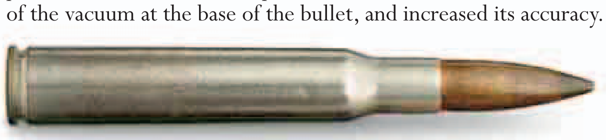
In the late 19th century, rifle bullets acquired a sharply pointed nose and a taper toward the tail. The shape minimized air resistance in flight, which almost doubled their effective range and improved their accuracy. In these examples, both velocity and energy are measured at the muzzle. The heavier the bullet and the higher its velocity, the greater is its energy.

8 × 58MM KRAG (1889)
This option for the Krag-Jørgensen rifle was adopted by the Danish Army. This 195-grain (12.7-g) bullet had a muzzle velocity of 2,525ft (770m) per second.

7.7 × 56MM JAPANESE (1889)
This fully rimmed cartridge—in which the rim was significantly wider than than the base of the cartridge—was used by the Arisaka rifle. It had a 175-grain (11.35g) bullet and a muzzle velocity of 2,350ft (716.3m) per second.

7.62 × 54MM RUSSIAN (1891)
This "3-line" cartridge was loaded with a 150-grain (9.65-g) bullet that left the muzzle at 2,855ft (870m) per second. The "line" is a caliber measure approximating one-tenth of an inch.

7.92 × 57MM MAUSER (1905)
Also called the SmK cartridge, this was loaded with a steel-jacketed 177-grain (11.5-g) bullet that left the muzzle at 2,745 ft (836.6m) per second. The boat-tail (tapered end) of the bullet reduced the size of the vacuum at the base of the bullet, and increased its accuracy.



.30IN-06 SPRINGFIELD (1906)
The .30in-06 remained in US service from 1906 until 1954. Its 152-grain (9.85-g) bullet left the muzzle at 2,910ft (887m) per second, with 2,820ft-lb (3,823J) of energy.

.470IN NITRO EXPRESS (1907)
"Nitro" refers to the propellant, while "Express" refers to the bullet, which was first produced in 1907. The bullet is hollow at the tip—on hitting the target, the bullet expands, reducing its penetration but increasing the tissue damage. Muzzle velocity of the bullet is 2,150ft (655.3m) per second, with 5,130ft-lb (6,955J) of energy.

7.7 × 56MM ITALIAN (1910)
The Italian 7.7mm cartridge had a 173-grain (11.25-g) bullet and a small charge with a muzzle velocity of 2,035ft (620.3m) per second.

.303IN MKVII (1910)
This version of the Lee-Enfield cartridge, with a 180-grain (11.66-g) bullet, had a muzzle velocity of 2,460ft (804.6m) per second and 2,420ft-lb (3,281J) of energy.

.50IN BROWNING /12.7MM M2 (1916/17)
Developed for the M2 machine-gun and adopted as a rifle round, this cartridge has a 710-grain (46-g) bullet and a muzzle velocity of 2,800ft (853.4m) per second.

.22IN HORNET (1920s)
One of very few high-velocity miniature rounds, the .22in Hornet was developed in the 1920s. Its 45-grain (2.9-g) bullet leaves the muzzle at 2,690ft (820m) per second.

7.92 × 33MM KURTZ (1938)
This was the first effective intermediate cartridge—less powerful than a typical battle rifle cartridge, such as the 7.62 × 54mm Russian, but significantly more powerful than pistol cartridges. It was developed in Nazi Germany and was copied by the Soviet Union in slightly smaller dimensions. It had a range of around 1,950ft (595m).

.257IN WEATHERBY MAGNUM (1944)
This is loaded with an 87-grain (5.31-g) "varmint" bullet—for rifles used to shoot small mammals, such as rodents. The cartridge achieves a muzzle velocity of 3,825ft (1165.8m) per second and delivers 2,826ft-lb (3,832J) of energy.